



Instagram & Wellbeing

A Public / Social Data Analysis

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Problem Background

Instagram is used by more than two billion people every month, which makes it a major part of how people socialise, relax, and spend their free time. There is growing public concern that heavy use is linked to higher stress and lower wellbeing. The objective of this project is to check whether the data actually supports that concern, and to describe how usage relates to stress, happiness, sleep, and exercise.

Dataset

Instagram Usage & Lifestyle Dataset (Kaggle)

The full dataset is public on Kaggle with over 1.5 million rows and 57 columns. It combines Instagram usage metrics with lifestyle, health, and self-reported wellbeing indicators for each user.

Key variables: daily active minutes, sessions, reels watched, passive consumption, perceived stress, happiness, sleep, and exercise.

Research Questions

- RQ1 Is there a link between heavy Instagram use and higher stress?
- RQ2 Does more usage relate to lower self-reported happiness?
- RQ3 Are wellbeing outcomes different across usage levels?
- RQ4 Does heavy usage connect to less sleep or less exercise?
- RQ5 Does reel consumption differ across age groups?

Analysis Process

01 Data Cleaning

- Remove missing values
- Handle outliers (IQR rule)
- Keep real heavy-usage values

02 EDA

- Visualise distributions
- Compare across usage tiers
- Find patterns and trends

03 Modelling

- Predict above-average stress
- Two classifiers compared
- Check the main drivers

Data Quality

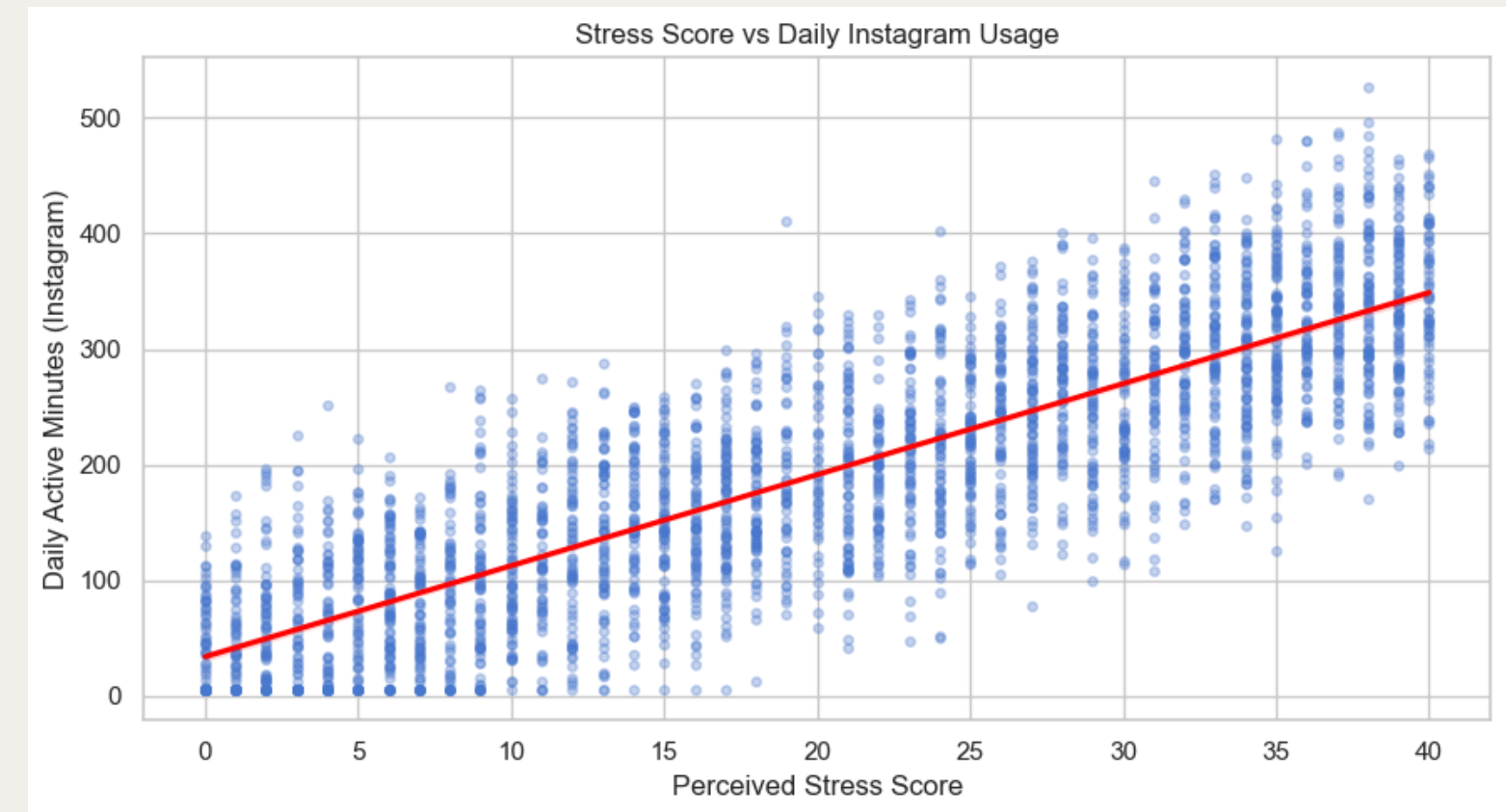
We compared the raw 50,000-row sample against the cleaned working set, to measure improvement:

- Missing values: 0 → 0 (nothing to fix)
- Duplicate rows: 0 → 0 (nothing to remove)
- IQR outliers on physiological columns: 1,631 → 60

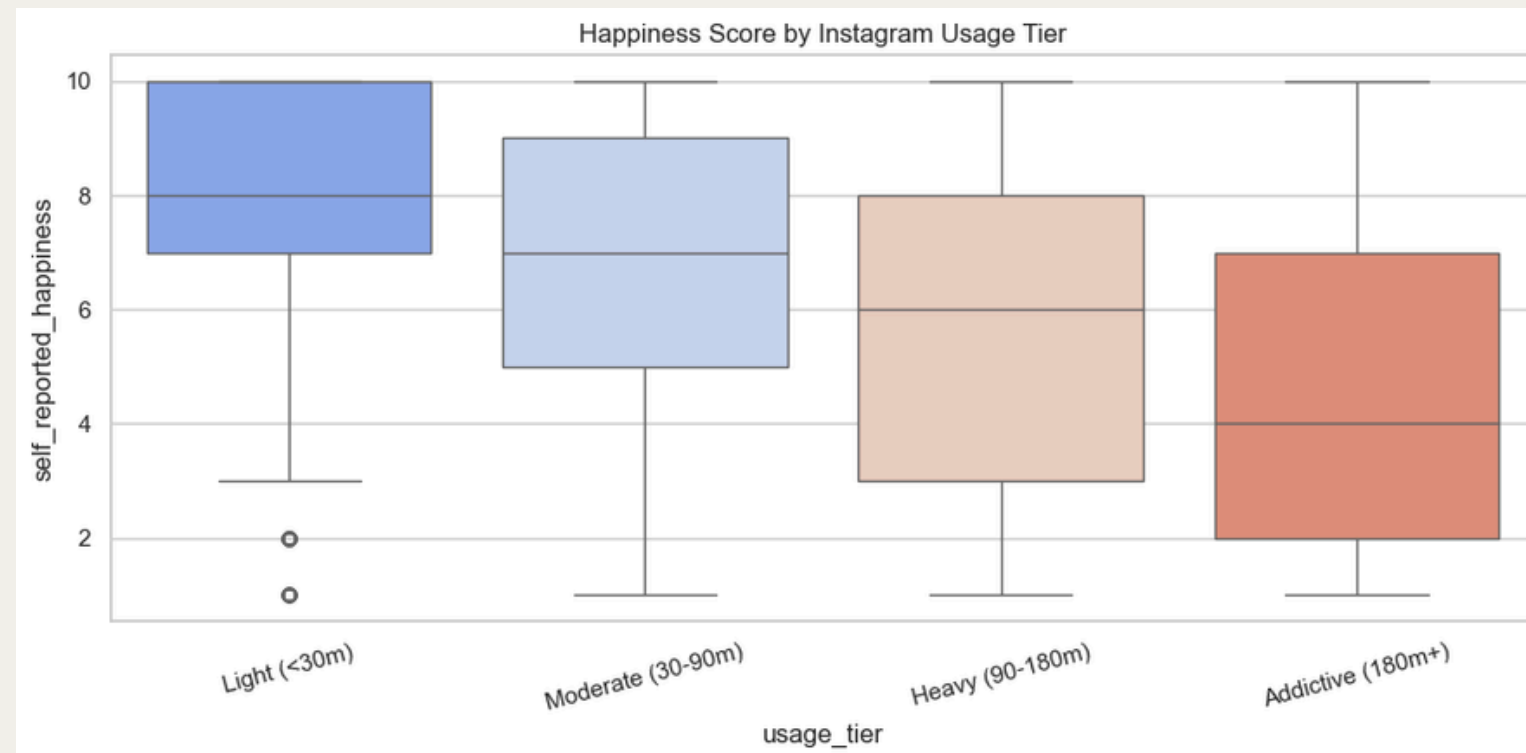
In total we dropped 1,646 rows (about 3.3%), keeping 48,354 rows — a small, careful, conservative reduction.

Key Findings #1

Daily Instagram minutes and stress move together very strongly, correlating at about $r = +0.84$, the largest effect in the whole dataset. As usage climbs, stress climbs with it. This directly answers RQ1: heavy use is clearly associated with higher stress.



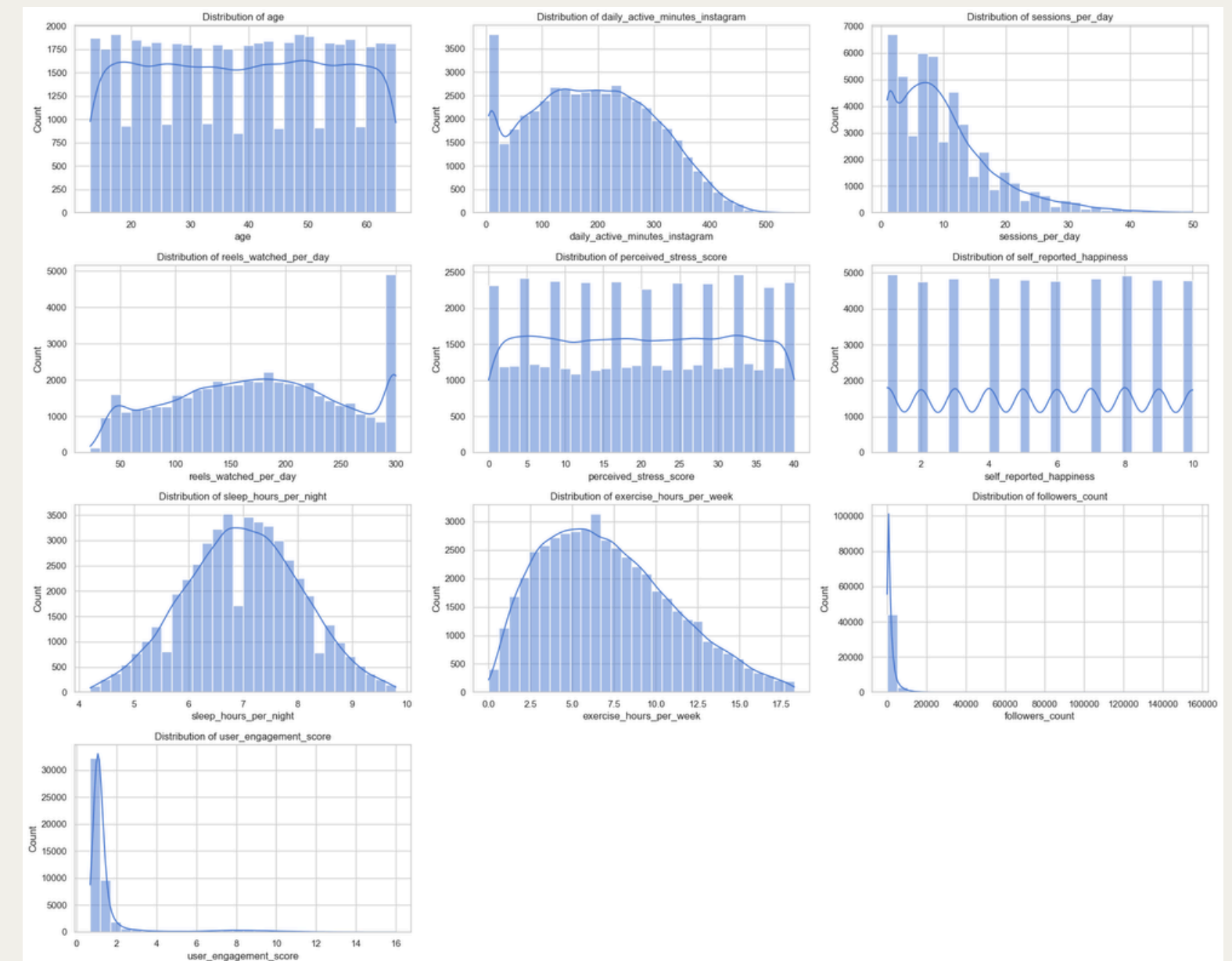
Key Findings #2



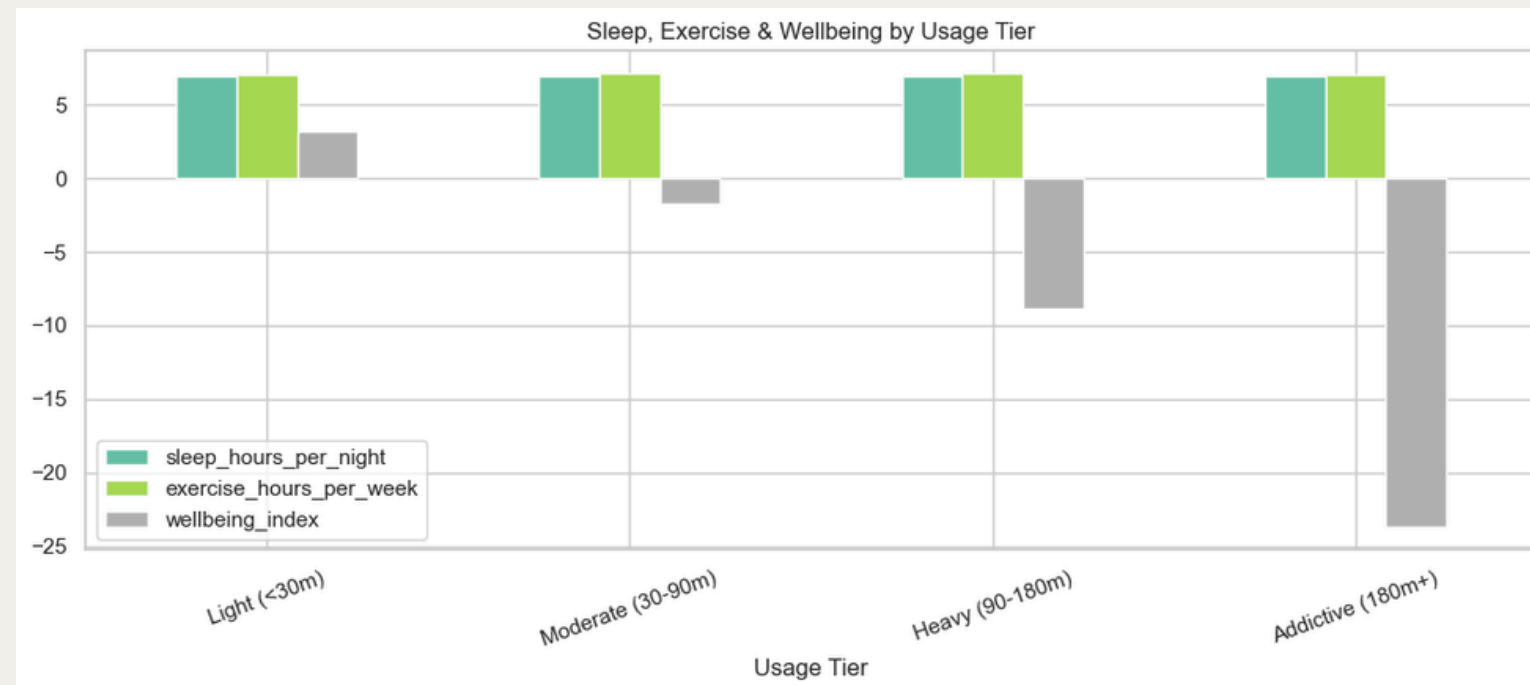
Median happiness falls steadily as usage rises, dropping from about 8 in the Light tier down to about 4 in the Addictive tier. The decline is consistent across every tier, not just at the extremes. This answers RQ2: more usage lines up with lower happiness.

Key Findings #3

The typical user spends about 188 minutes (over three hours) on Instagram every day. Most numerical variables are clean and roughly bell-shaped after cleaning, with no broken scales, so the data is reliable enough to analyse.



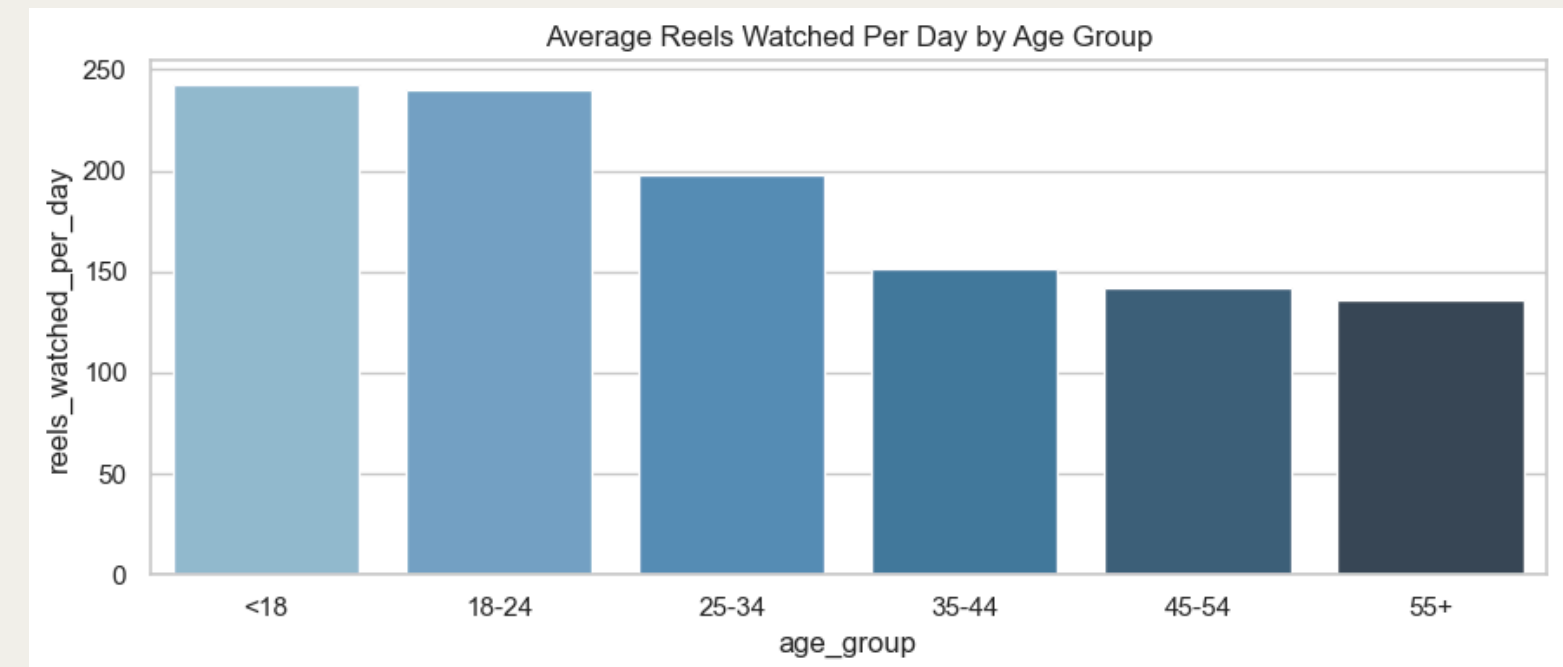
Key Findings #4



Average sleep stays around seven hours a night and exercise stays flat across every usage tier, while the wellbeing index still drops sharply. So heavy usage is NOT linked to less sleep or exercise which is a useful null result for RQ4. The wellbeing story is about stress and mood, not lifestyle.

Key Findings #5

Average daily reels watched decline smoothly with age, from the youngest users down to the oldest. Under-25s watch far more reels than over-45s. This answers RQ5 and points to younger cohorts as the place to focus any reel-related intervention.

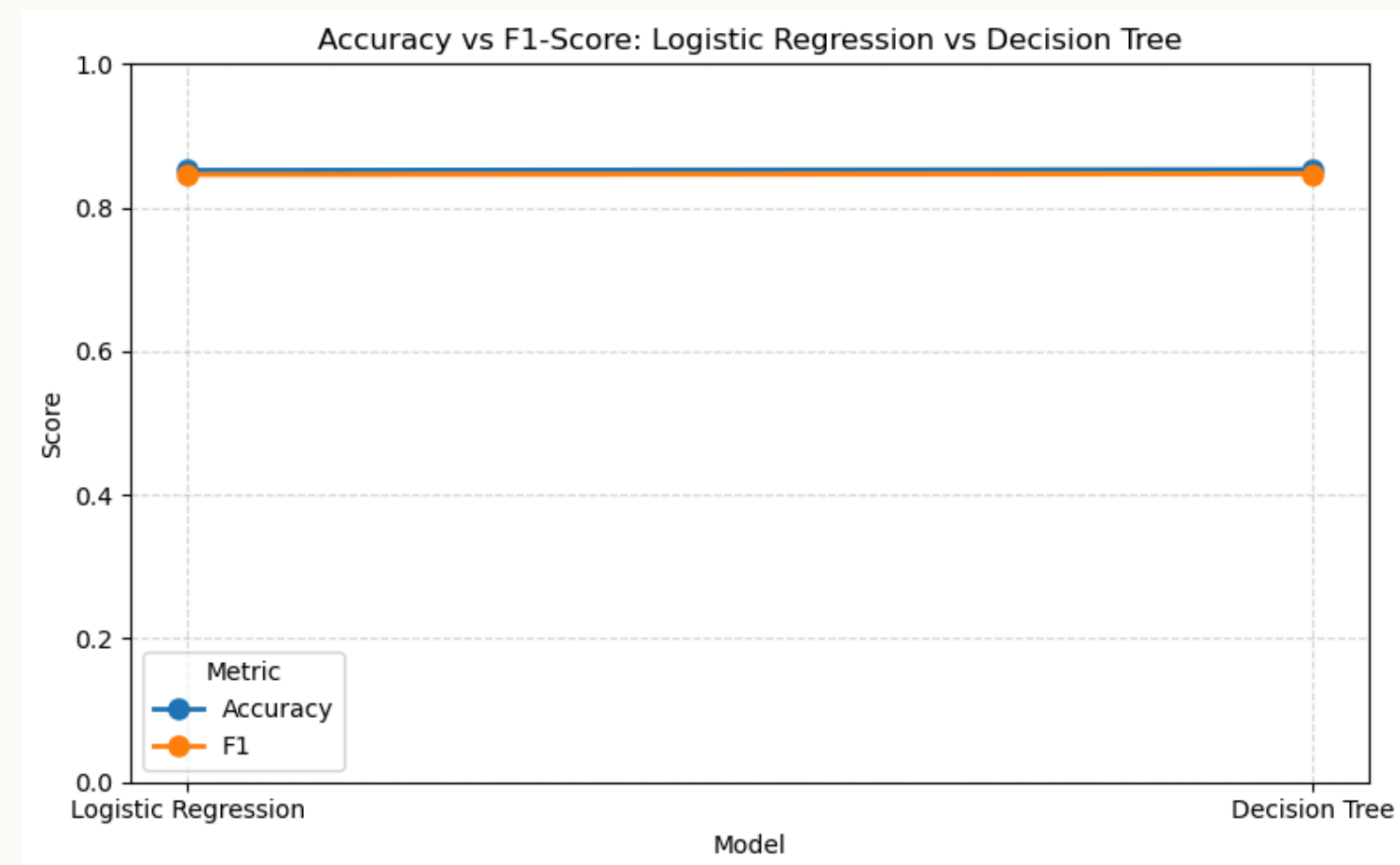


Data Modelling

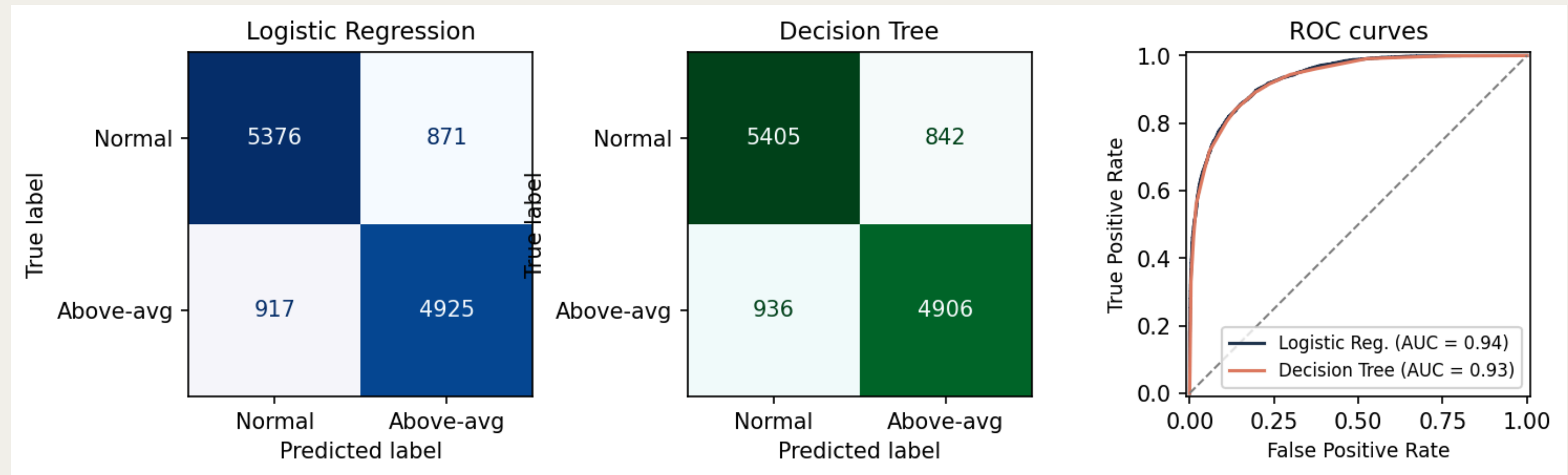
- **Algorithms: Logistic Regression + Decision Tree**
Two classifiers are trained and compared on the same task, so we can check whether they agree instead of trusting one model.
- **Target: above-average stress**
A user is flagged if their perceived stress score is above the median. The split is almost 50/50, so the task is balanced.
- **Features: four usage behaviours**
Daily active minutes, passive consumption, sessions per day, and reels watched per day — the behaviours we can actually act on.

Data Modelling

- **Train / test split: 75% train, 25% test**
The Decision Tree uses `max_depth = 4` and a fixed `random_state`, so the comparison is fair and reproducible.
- **Evaluation metrics**
Accuracy, precision, recall, F1, confusion matrix, and ROC AUC, judged on the held-out test set the models never saw.
- **Results: both models score about 85%**
Accuracy 85.2% vs 85.3%; ROC AUC 0.935 vs 0.933. Both agree that daily active minutes is the main driver of stress (the tree puts ~97% of its importance there).



Model Results



Confusion matrices and ROC curves — both models perform almost identically.

Business Recommendations

1 Show a gentle "time spent today" reminder past about 90 minutes of use, and a stronger one past 180 minutes.

2 Nudge heavy users away from passive scrolling toward active, interactive use — passive consumption is the strongest driver of stress.

3 Drop sleep and exercise messaging. Measure wellbeing success on stress and mood instead, where the real link is.

4 Prioritise the Addictive tier (about 52% of users, the worst wellbeing) and target reel-pacing prompts at younger users.

Limitations & Risks

- The data is cross-sectional, a single snapshot, so it shows association, not cause.
- Wellbeing scores are self-reported, and the data covers one platform at one point in time.
- These correlations and predictions are not proof , an intervention may help less than the numbers suggest, so test with an A/B trial first.
- Segment users by behaviour, not by age or gender, to avoid unfair targeting.

Conclusion

- Usage tracks wellbeing: it correlates with stress (+0.84) and happiness (-0.37), strongest in the heaviest-usage tier.
- Sleep and exercise are NOT linked to usage, stress and mood are where the story is.
- Two different models agree that daily usage time is the main driver of above-average stress.
- The findings support gentle, behaviour-based digital-wellbeing nudges, validated before rollout.



Thank You

Questions & Discussion